



The University of Lahore

Department of CS & IT

Assignment 04

Instructor Name: Miss Iqra Adnan

Course Code: CS09202 | 11

Course Title: Discrete Structures

Assigned Date: 03-01-2025

Due Date: 10-01-2025

NAME: Mian Daniyal Hassan

SAP ID: 70150006

Question 1: Prove that $1^2 + 3^2 + 5^2 + \dots + (2n + 1)^2 = (n + 1)(2n + 1)(2n + 3)/3$ whenever n is a nonnegative integer.

Answer :

Base case ($n = 0$):

- L.H.S: $(2(0) + 1)^2 = 1^2$
- R.H.S: $(0 + 1)(2(0) + 1)(2(0) + 3)/3 = (1)(1)(3)/3 = 1$

Inductive Step:

$$n = k: 1^2 + 3^2 + 5^2 + \dots + (2k + 1)^2 = (k + 1)(2k + 1)(2k + 3)/3$$

L.H.S

$$\begin{aligned} & 1^2 + 3^2 + 5^2 + \dots + (2k + 1)^2 + (2(k+1) + 1)^2 \\ &= (k + 1)(2k + 1)(2k + 3)/3 + (2k + 3)^2 \\ &= [(k + 1)(2k + 1)(2k + 3) + 3(2k + 3)^2]/3 \\ &= [(k + 1)(2k + 1)(2k + 3) + 3(2k + 3)^2]/3 \\ &= [(k + 2)(2k + 3)(2k + 5)]/3 \\ &= ((k + 1) + 1)(2(k + 1) + 1)(2(k + 1) + 3)/3 \end{aligned}$$

Question # 2: Find the sum of the first 18 terms of the geometric sequence 2, 6, 18, 54,...

Solution:

- First term (a) = 2
- Common ratio (r) = 3
- Number of terms (n) = 18

$$S_n = a(1-r^n)/(1-r)$$

$$S_{18} = 2(1-3^{18})/(1-3)$$

$$= 2(1-387420489)/-2$$

$$= 2(387420488)/2$$

$$= 387420488$$

Question # 3: Determine whether the sequence $\{a_n\}$, where $a_n = 2^n$ for every nonnegative integer n , is a solution of the recurrence relation that satisfies the recurrence relation

$$a_n = 2a_{n-1} - a_{n-2} \quad \text{for } n = 2, 3, 4, \dots$$

Solution:

- L.H.S: $a_n = 2^n$
- R.H.S: $2a_{n-1} - a_{n-2}$
- $= 2(2^{n-1}) - 2^{n-2}$
- $= 2^n - 2^{n-2}$
- $= 2^n - 2^{n/4}$
- $= 2^n(1 - 1/4)$
- $= 2^n(3/4) \neq 2^n$

Hence, $a_n = 2^n$ is NOT a solution to the recurrence relation.

Question # 4:

$$\sum_{j=0}^n \left(-\frac{1}{2}\right)^j = \frac{2^{n+1} + (-1)^n}{3 \cdot 2^n}$$

whenever n is a nonnegative integer.

Answer:

base case ($n = 0$):

- L.H.S: $(-1/2)^0 = 1$
- R.H.S $(2^{0+1} + (-1)^0)/(3 \cdot 2^0) = (2 + 1)/3 = 1$

- Base case proved

$n = k$ (Inductive Step):

$$\begin{aligned}
 & \sum_{j=0}^k (-1/2)^j \\
 &= (2^{k+1} + (-1)^k) / (3 \cdot 2^k) \\
 \text{L.H.S} &= \sum_{j=0}^{k+1} (-1/2)^j \\
 &= [\sum_{j=0}^k (-1/2)^j] + (-1/2)^{k+1} \\
 &= (2^{k+1} + (-1)^k) / (3 \cdot 2^k) + (-1/2)^{k+1} \\
 &= (2^{k+1} + (-1)^k) / (3 \cdot 2^k) + (-1)^{k+1} / 2^{k+1} \\
 &= (2^{k+2} + (-1)^{k+1}) / (3 \cdot 2^{k+1}) + (-1)^{k+1} / 2^{k+1} \\
 &= (2^{k+2} + (-1)^{k+1} \cdot 3) / (3 \cdot 2^{k+1}) \\
 &= (2^{k+2} + (-1)^{k+1}) / (3 \cdot 2^{k+1})
 \end{aligned}$$

Question # 5: Let $P(n)$ be the statement that $1^3 + 2^3 + \dots + n^3 = (n(n+1)/2)^2$ for the positive integer n .

a) What is the statement $P(1)$?

$P(1)$ states that: $1^3 = (1(1+1)/2)^2$ In other words, $1^3 = (1 \times 2/2)^2 = 1^2 = 1$

b) Show that $P(1)$ is true, completing the basis step of the proof.

L.H.S: $1^3 = 1$

R.H.S: $(1(1+1)/2)^2 = (1 \times 2/2)^2 = 1^2 = 1$

Since L.H.S = R.H.S $P(1)$ is true

c) What is the inductive hypothesis?

Assume $P(k)$ is true for some positive integer k That is: $1^3 + 2^3 + \dots + k^3 = (k(k+1)/2)^2$

d) What do you need to prove in the inductive step?

We need to prove $P(k+1)$ is true, assuming $P(k)$ is true.

Which is, we need to prove: $1^3 + 2^3 + \dots + k^3 + (k+1)^3 = ((k+1)(k+2)/2)^2$

e) Complete the inductive step, identifying where you use the inductive hypothesis.

L.H.S of $P(k+1)$:

$$1^3 + 2^3 + \dots + k^3 + (k+1)^3$$

$$= (k(k+1)/2)^2 + (k+1)^3$$

$$= k^2(k+1)^2/4 + (k+1)^3$$

$$= (k+1)^2[k^2/4 + (k+1)]$$

$$= (k+1)^2[k^2/4 + k + 1]$$

$$= (k+1)^2[(k^2 + 4k + 4)/4]$$

$$= (k+1)^2(k+2)^2/4$$

$$= ((k+1)(k+2)/2)^2$$

R.H.S P(k+1)

f) Explain why these steps show that this formula is true whenever n is a positive integer

Because:

1. We proved it's true for $n = 1$ (base case)
2. We proved that if it's true for any positive integer k , then it's also true for $k+1$ (inductive step)
3. By the principle of mathematical induction, the statement is true for all positive integers n